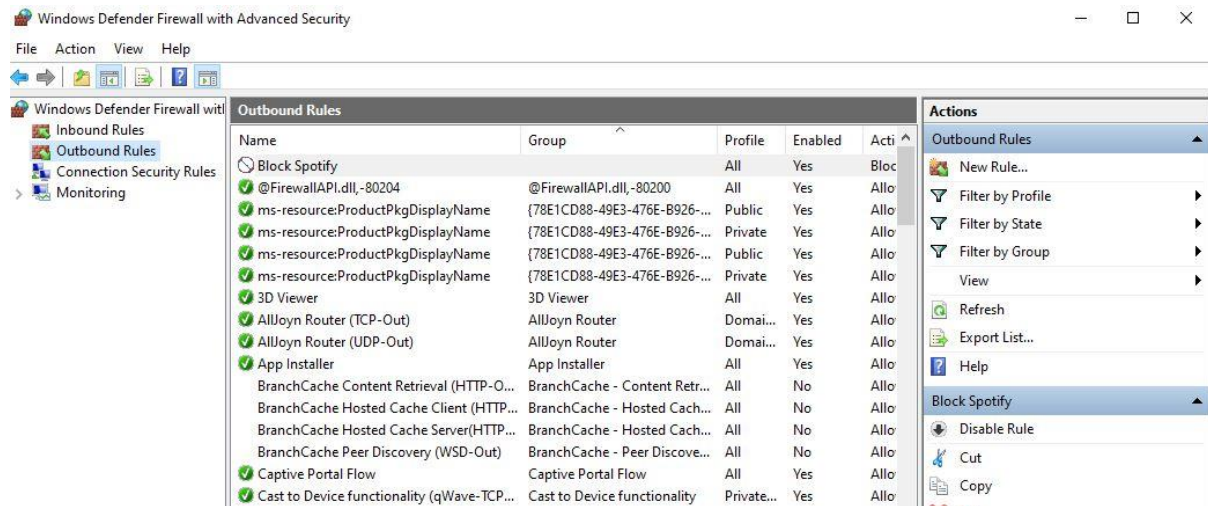
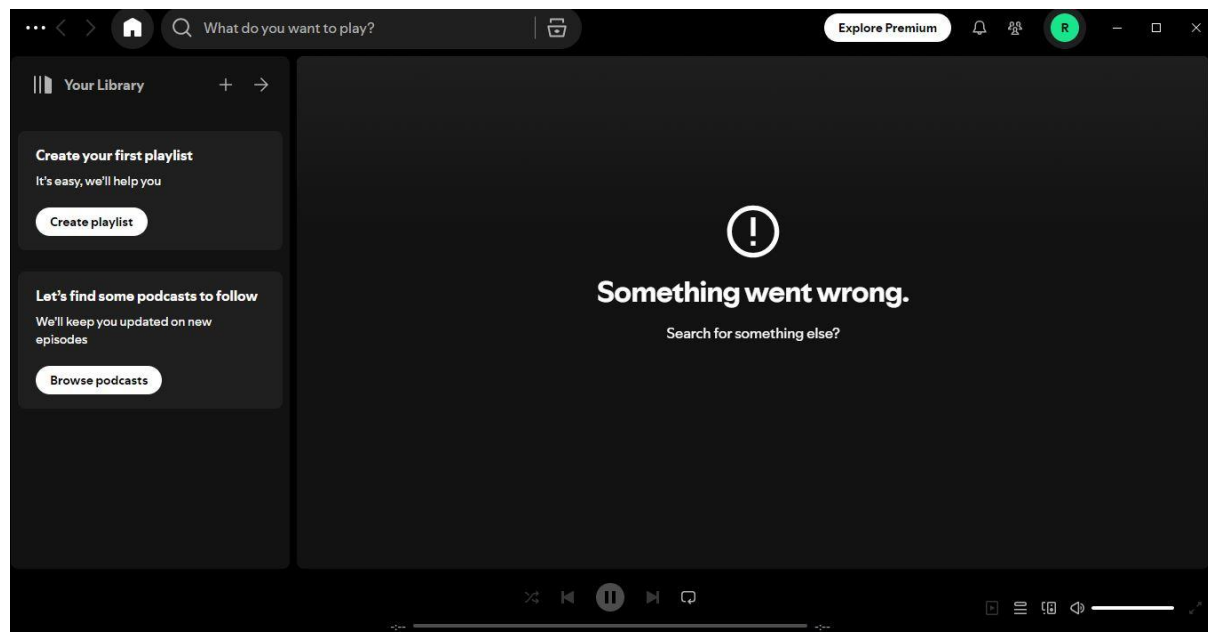


# 1. Open Windows Defender Firewall settings on your computer and create a new outbound rule to block all traffic from the Spotify desktop app.

- Windows defender Firewall Page



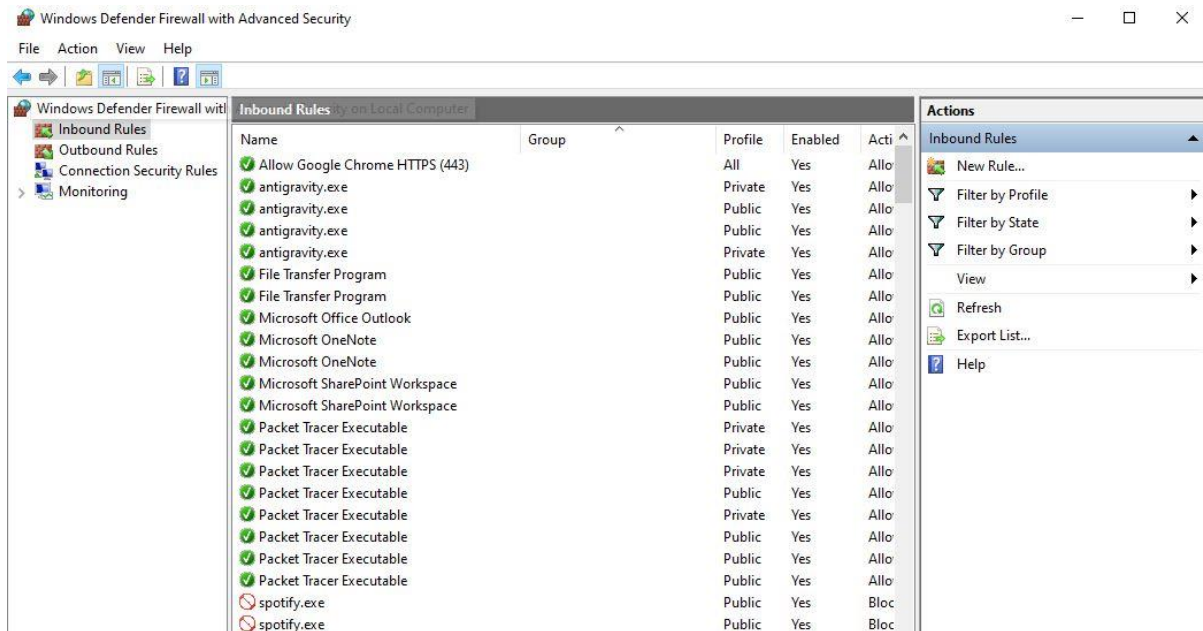
- Spotify Page



- A new outbound firewall rule named "Block Spotify" was created for Spotify.exe. After enabling the rule, Spotify was unable to access the internet, confirming that all outbound traffic from the application was successfully blocked.

## 2. Set up an inbound rule in Windows Defender Firewall to allow incoming connections only for Google Chrome on port 443 (HTTPS).

- Here, I have setup an inbound in windows defender firewall which allows incoming connections for Google Chrome on HTTP 443 port.



## 3. Document, step-by-step, how you would block all traffic from a specific IP address (for example, 123.45.67.89) using Windows Defender Firewall.

- Blocking all traffic from a specific IP address using Windows defender firewall.

### Step 1 Open Windows Defender Firewall

- 1) Press Windows + R
- 2) Type wf.msc and press enter.
- 3) The windows defender firewall with advanced security window will open.

### Step 2 Create a New Inbound Rule

- 1) In the left panel, click inbound rules.
- 2) In the right panel, click new rule.

### Step 3 Select Rule Type

- 1) Choose Custom.
- 2) Click Next.

#### **Step 4 Apply to All Programs**

- 1) Select All Programs.
- 2) Click Next.

#### **Step 5 Protocol and Ports**

- 1) Leave the default setting for protocol and ports.
- 2) Click Next.

#### **Step 6 Configuration to Block Specific IP**

- 1) In the Scope tab:
  - Under Which remote IP addresses does this rule apply to? Select These IP addresses.
  - Click Add.
  - Enter 123.45.67.89
  - Click OK
- 2) Click Next.

#### **Step 7 Choose the Action**

- 1) Select Block the connection.
- 2) Click Next.

#### **Step 8 Select Profiles**

- 1) Check all three profiles :
  - Domain
  - Private
  - Public
- 2) Click Next.

#### **Step 9 Name the Rule**

- 1) Enter the rule name: BLOCK IP 123.45.67.89 (Inbound)
- 2) Optionally add a description.
- 3) Click Finish.

### **Outbound Rule**

1. Select Outbound Rules.
2. Click New Rule.
3. Choose Custom.
4. Select all programs.
5. Leave any protocol and ports.
6. In the scope tab, add the remote IP address 123.45.67.89
7. Choose Block the connection.
8. Apply the rule to Domain, Private Public profiles.
9. Name the rule: BLOCK IP 123.45.67.89 (Outbound)
10. Click Finish.

- 4. Compare hardware firewalls (like those used in large offices) and software firewalls (like Windows Defender) by listing 3 key differences and a use case for each.**

| <b>Feature</b>   | <b>Hardware Firewall</b>                                          | <b>Software Firewall</b>                                |
|------------------|-------------------------------------------------------------------|---------------------------------------------------------|
| Location         | A physical device installed between the internet and the network. | Software installed on an individual computer or server. |
| Protection Scope | Protects the entire network and all connected devices.            | Protects only the device on which it is installed.      |
| Management       | Centrally managed by network administrators.                      | Managed on each computer separately.                    |